AUGMENTED DECISION

Regularization and variable selection

Ridge, lasso, and the same warehouse prediction task

Lecture 4

What regularization can improve

The full OLS model fits $p = 24$ slopes from only $n = 60$ training observations. Correlated columns can make individual coefficients sensitive to noise, even though $n \geq p$.

- Ridge stabilizes the fit by shrinking coefficients.
- Lasso can shrink some coefficients all the way to zero.

Compare prediction, model size, and variable stability on the same task. A penalty does not guarantee that every criterion improves.

Ridge regression: objective and closed form

For $\lambda > 0$, define

$$\widehat{\beta}_{\lambda}^{\mathrm{ridge}} = \arg\min_{b} \left\{ \frac{1}{2n} \|\mathbf{Y} - \mathbf{X}b\|_2^2 + \frac{\lambda}{2} \|b\|_2^2 \right\}.$$

Recall that $G = \mathbf{X}^\top \mathbf{X}/n$, and differentiating gives

$$\widehat{\beta}_{\lambda}^{\mathrm{ridge}} = (G + \lambda I_p)^{-1} \frac{\mathbf{X}^\top \mathbf{Y}}{n} = (\mathbf{X}^\top \mathbf{X} + n\lambda I_p)^{-1} \mathbf{X}^\top \mathbf{Y}.$$

The objective is strictly convex for $\lambda > 0$, even when $p > n$.

Ridge bias and variance

$$\mathbb{E}\hat{\beta}_\lambda^{\mathrm{ridge}} - \beta^* = -\lambda(G + \lambda I_p)^{-1}\beta^*,$$
$$\mathrm{Cov}(\hat{\beta}_\lambda^{\mathrm{ridge}}) = \frac{\sigma^2}{n}(G + \lambda I_p)^{-1} G (G + \lambda I_p)^{-1}.$$

If $G = U\,\mathrm{diag}(t_1, \ldots, t_p)U^\top$ and $b = U^\top \beta^*$,

$$\mathcal{R}_2 = \sum_{j=1}^{p} \frac{\lambda^2 b_j^2}{(t_j + \lambda)^2} + \frac{\sigma^2}{n} \sum_{j=1}^{p} \frac{t_j}{(t_j + \lambda)^2}.$$

Shrinkage reduces variance while introducing parameter-dependent bias.

The prediction-risk formula has an extra factor t_j in each summand. [RT, Ridge notes]

Ridge on an orthogonal Gaussian model

If $G = I_p$, then $\widehat{\beta}_\lambda = W/(1+\lambda)$ and

$$\mathcal{R}_2(\widehat{\beta}_\lambda, \beta^*) = \frac{\lambda^2 \|\beta^*\|_2^2 + p\tau^2}{(1+\lambda)^2}.$$

On the ball $\|\beta^*\|_2 \le B$, $B > 0$, the best fixed ridge tuning is

$$\lambda_* = \frac{p\tau^2}{B^2}, \qquad \sup_{\|\beta\|_2 \le B} \mathcal{R}_2(\widehat{\beta}_{\lambda_*}, \beta) = \frac{B^2 p\tau^2}{B^2 + p\tau^2}.$$

This minimizes worst-case risk within the scalar ridge family. The ball's full minimax rate is $\asymp \min\{B^2, p\tau^2\}$.

A parameter set changes the meaning of optimality

For fixed $\lambda > 0$ in the orthogonal model,

$$\sup_{\beta^* \in \mathbb{R}^p} \mathcal{R}_2(\widehat{\beta}_\lambda^{\text{ridge}}, \beta^*) = \infty.$$

The bias term grows without bound as $\|\beta^*\|_2 \to \infty$.

- Least squares has constant risk $p\tau^2$ on the unrestricted space.
- Ridge can improve risk on bounded parameter sets or at particular parameters.
- Sparsity without an amplitude bound still allows arbitrarily large coefficients.

Lasso: objective and optimality conditions

For $\lambda > 0$,

$$\widehat{\beta}_{\lambda}^{\text{lasso}} \in \arg\min_{b} \left\{ \frac{1}{2n} \left\| \mathbf{Y} - \mathbf{X} b \right\|_2^2 + \lambda \left\| b \right\|_1 \right\}.$$

The KKT condition is

$$\frac{\mathbf{X}^\top (\mathbf{Y} - \mathbf{X}\widehat{\beta})}{n} = \lambda z, \qquad z_j \in \begin{cases} \{\operatorname{sign}(\widehat{\beta}_j)\}, & \widehat{\beta}_j \neq 0, \\ [-1, 1], & \widehat{\beta}_j = 0. \end{cases}$$

These are the subgradients of absolute value. Coefficient minimizers can be nonunique for a singular design; the bounds below apply to every minimizer.

Orthogonal lasso is soft thresholding

With $G = I_p$, the objective separates into scalar problems:

$$\widehat{\beta}_j \in \arg\min_b \{\tfrac{1}{2}(W_j - b)^2 + \lambda|b|\}.$$

The solution is

$$\widehat{\beta}_j = \mathrm{sign}(W_j)(|W_j| - \lambda)_+, \qquad (t)_+ = \max\{t, 0\}.$$

- If $|W_j| \le \lambda$, the coordinate becomes exactly zero.
- Otherwise, it moves toward zero by λ.
- Ridge instead scales every coordinate by $1/(1+\lambda)$.

A small shrinkage calculation

For the orthogonal statistic $W = (3,\ 0.8,\ -0.2)^\top$, take $\lambda = 1$.

$$\widehat{\beta}^{\text{ridge}} = (1.5,\ 0.4,\ -0.1)^\top, \qquad \widehat{\beta}^{\text{lasso}} = (2,\ 0,\ 0)^\top.$$

- Ridge retains all three coordinates with smaller magnitudes.
- Lasso selects only the first coordinate in this example.
- For lasso, $W - \widehat{\beta} = (1, 0.8, -0.2)^\top$: the zero-coordinate scores lie inside $[-\lambda, \lambda]$.

The same numeric λ illustrates the formulas. It is not a rule for tuning ridge and lasso to equal penalty values.

The Gaussian score event and tuning level

Assume normalized columns $\|\mathbf{X}_j\|_2 \le \sqrt{n}$ and Gaussian noise. Each coordinate of $\mathbf{X}^\top \varepsilon / n$ has variance at most σ^2/n.

$$\mathbb{P}\left\{ \left\| \frac{\mathbf{X}^\top \varepsilon}{n} \right\|_\infty > t \right\} \le 2p \exp\left(-\frac{n t^2}{2\sigma^2} \right).$$

Choose

$$\lambda = 2\sigma \sqrt{\frac{2 \log(2p/\alpha)}{n}}.$$

Then $\mathcal{A}_\lambda = \{ \left\| \mathbf{X}^\top \varepsilon / n \right\|_\infty \le \lambda/2 \}$ has probability at least $1 - \alpha$.

The lasso basic inequality

Let $\Delta = \widehat{\beta} - \beta^*$ and $S = \text{supp}(\beta^*)$. Comparing the objective at $\widehat{\beta}$ and β^* gives

$$\frac{1}{2n} \|\mathbf{X}\Delta\|_2^2 \leq \frac{\varepsilon^\top \mathbf{X}\Delta}{n} + \lambda(\|\beta^*\|_1 - \|\widehat{\beta}\|_1).$$

On $\mathcal{A}_\lambda$,

$$\frac{1}{2n} \|\mathbf{X}\Delta\|_2^2 \leq \frac{3\lambda}{2} \|\Delta_S\|_1 - \frac{\lambda}{2} \|\Delta_{S^c}\|_1 .$$

Use Hölder's inequality and $\|\beta^*\|_1 - \|\beta^* + \Delta\|_1 \leq \|\Delta_S\|_1 - \|\Delta_{S^c}\|_1$.

The error lies in a sparse cone

Multiplying the basic inequality by two yields

$$\underbrace{\frac{1}{n}\left\|\mathbf{X}\Delta\right\|_2^2}_{d^2} + \lambda \left\|\Delta_{S^c}\right\|_1 \leq 3\lambda \left\|\Delta_S\right\|_1 .$$

Since $d^2 \geq 0$,

$$\Delta \in \mathcal{C}(S, 3) = \{u : \left\|u_{S^c}\right\|_1 \leq 3 \left\|u_S\right\|_1\}.$$

Also $d^2 \leq 3\lambda\sqrt{s}\left\|\Delta\right\|_2$ when $|S| \leq s$.

The error need not be sparse, but its mass is controlled relative to the true support.

A restricted-eigenvalue assumption

Assume a constant $\kappa > 0$ such that, for every $|S| \leq s$,

$$\frac{1}{n} \|\mathbf{X}u\|_2^2 \geq \kappa^2 \|u\|_2^2 \quad \text{for all } u \in \mathcal{C}(S, 3).$$

- This lower bound is required only on sparse cones.
- A global positive minimum eigenvalue is impossible when $p > n$.
- The displayed condition can still hold in suitable sparse problems.

We use the full-$\|u\|_2$ cone version of RE. Other definitions use $\|u_S\|_2$ and do not give identical coefficient bounds. [RH, Section 10.3]

Lasso prediction and coefficient bounds

On $\mathcal{A}_\lambda$, let $d = \|\mathbf{X}\Delta\|_2 / \sqrt{n}$. Under the stated RE condition,

$$d^2 \le 3\lambda\sqrt{s}\,\|\Delta\|_2 \le \frac{3\lambda\sqrt{s}}{\kappa} d.$$

Thus, with probability at least $1 - \alpha$,

$$\boxed{\mathcal{L}_{\mathbf{X}} \le \frac{9s\lambda^2}{\kappa^2}}, \qquad \boxed{\mathcal{L}_2 \le \frac{9s\lambda^2}{\kappa^4}}.$$

Also $\|\Delta\|_1 \le 12s\lambda/\kappa^2$.

The case $d = 0$ follows directly from RE. Otherwise divide by d.

Comparison with the sparse minimax benchmark

The chosen λ gives the high-probability scales

$$\mathcal{L}_{\mathbf{x}} \lesssim \frac{\sigma^2 s \log(2p/\alpha)}{n\kappa^2}, \qquad \mathcal{L}_2 \lesssim \frac{\sigma^2 s \log(2p/\alpha)}{n\kappa^4}.$$

The orthogonal sparse minimax benchmark is

$$\frac{\sigma^2 s \log(ep/s)}{n}.$$

The logarithms differ. With fixed confidence and $s \leq p^{1-\gamma}$, $\gamma > 0$, they are comparable up to constants depending on γ. Expected minimax risk and a high-probability loss bound are different statements.

Prediction guarantees do not imply support recovery

If two columns are identical, a signal on either column produces the same mean.

$$\mathbf{X}_1 = \mathbf{X}_2 \quad \Longrightarrow \quad \mathbf{X}(a, 0, \ldots)^\top = \mathbf{X}(0, a, \ldots)^\top.$$

- Prediction may be accurate while the selected variables are ambiguous.
- Exact support recovery needs further assumptions, including signal strength and suitable design structure.
- A nonzero fitted coefficient does not establish a causal effect.

Tuning must respect the split and the objective

- Fit the scaler and coefficients on the 60 training rows only.
- For each method, choose λ by the smallest validation MSE on the supplied grid.
- Compare the chosen ridge and lasso fits with the frozen Lecture 3 models.
- Keep these training fits for the final test comparison. No refitting on training plus validation is used in this lab.

The theoretical λ assumes known noise scale and a score event. A validation-selected λ does not automatically satisfy that theorem.

Use fresh evaluation for the procedure actually fitted.

From the mathematical penalty to Python

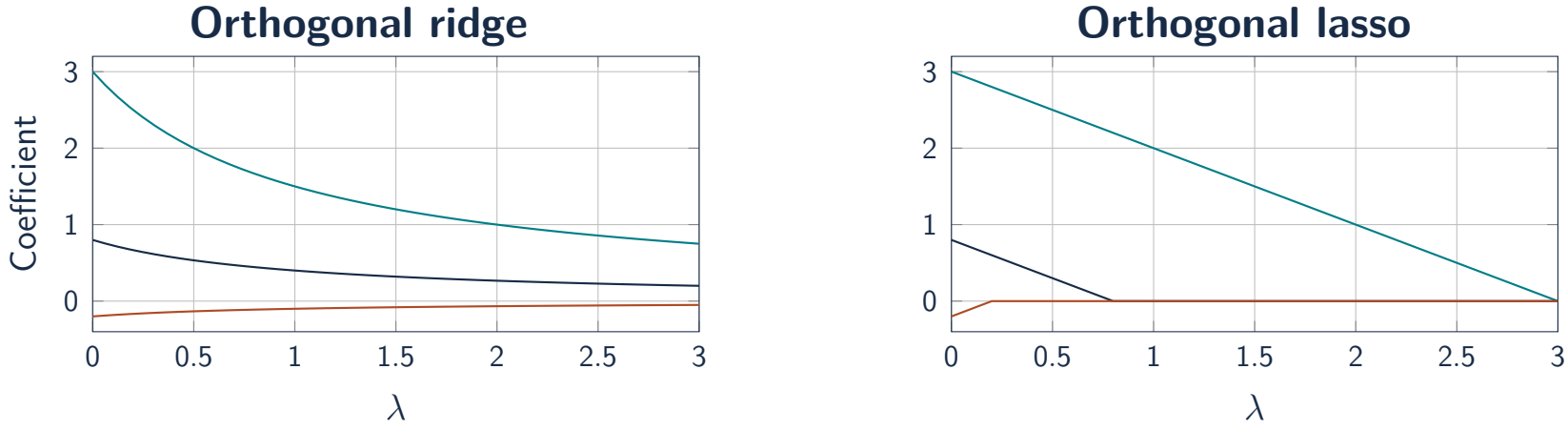
Center and scale using training rows, then work with $y_c = y - \bar{y}$ and $\widetilde{X}$.

Lecture objective	Matching scikit-learn call
$\mathrm{RSS}/(2n) + \lambda\|b\|_2^2/2$	`Ridge(alpha=n*lam, fit_intercept=False)`
$\mathrm{RSS}/(2n) + \lambda\|b\|_1$	`Lasso(alpha=lam, fit_intercept=False)`

$$\hat{\theta}_{\mathrm{ridge}} = (\widetilde{X}^\top \widetilde{X}/n + \lambda I)^{-1} \widetilde{X}^\top y_c/n.$$

Implement ridge with `numpy.linalg.solve`. Add $\bar{y}$ back to predictions. The library argument named `alpha` is a penalty, not the lecture's failure probability α.

Regularization paths show how coefficients change



The same $W = (3, 0.8, -0.2)$ as before. These are analytic orthogonal paths. Correlated-design lasso paths need not follow these coordinatewise formulas.

Selection stability is another piece of evidence

After choosing k or λ, repeatedly resample training rows and refit the procedure.

$$\widehat{\pi}_j = \frac{\text{number of bootstrap fits selecting variable } j}{\text{number of bootstrap fits}}.$$

- Keep k and λ fixed. Refit scaling inside each resample.
- Correlated measurements may exchange places across fits.
- Ridge generally has no exact zeros; a small coefficient is not automatically a selected-out variable.

Selection frequency is descriptive here. It is not a calibrated probability that a variable belongs to the true support.

Appendix: ridge prediction risk and effective dimension

With eigenvalues t_j of G and coordinates $b = U^\top \beta^*$,

$$\mathcal{R}_{\mathbf{X}}(\widehat{\beta}_\lambda^{\text{ridge}}, \beta^*) = \sum_j \frac{t_j \lambda^2 b_j^2}{(t_j + \lambda)^2} + \frac{\sigma^2}{n} \sum_j \frac{t_j^2}{(t_j + \lambda)^2}.$$

The smoother is $H_\lambda = \mathbf{X}(\mathbf{X}^\top \mathbf{X} + n\lambda I)^{-1} \mathbf{X}^\top$.

$$\text{df}(\lambda) = \operatorname{tr} H_\lambda = \sum_j \frac{t_j}{t_j + \lambda}, \qquad \text{variance term} = \frac{\sigma^2}{n} \operatorname{tr}(H_\lambda^2).$$

Effective degrees of freedom and the prediction variance involve different powers of the shrinkage factors.

Appendix: the minimax rate on a Euclidean ball

In the sequence model with $\|\beta\|_2 \le B$, the best fixed ridge rule gives

$$\mathfrak{M}(B) \le \frac{B^2 p\tau^2}{B^2 + p\tau^2} \le \min\{B^2, p\tau^2\}.$$

For a lower bound choose $\beta_j \in \{-u, u\}$, with

$$u = c \min\{B/\sqrt{p}, \tau\}.$$

The hypercube lies in the ball. Neighboring experiments have $\mathrm{KL} = 2u^2/\tau^2$, bounded by a small constant for suitable universal c. Testing each sign forces squared error of order u^2 on a constant fraction of coordinates. Therefore

$$\mathfrak{M}(B) \ge c' p u^2 \asymp \min\{B^2, p\tau^2\}.$$

Appendix: a lasso bound without restricted eigenvalues

On the same score event, basic optimality gives

$$\begin{aligned}\tfrac{1}{2}\mathcal{L}_{\mathbf{X}} &\leq \tfrac{\lambda}{2}\left\|\widehat{\beta}-\beta^*\right\|_1+\lambda(\|\beta^*\|_1-\left\|\widehat{\beta}\right\|_1)\\ &\leq \tfrac{3\lambda}{2}\|\beta^*\|_1-\tfrac{\lambda}{2}\left\|\widehat{\beta}\right\|_1.\end{aligned}$$

Consequently,

$$\mathcal{L}_{\mathbf{X}} \leq 3\lambda\|\beta^*\|_1.$$

This prediction bound needs column normalization and the score event, but no sparsity or RE condition. It does not imply coefficient recovery.

Tuning and the scope of a guarantee

- The lasso theorem uses a prescribed λ large enough for the score event.
- A cross-validated λ need not meet that threshold automatically.
- Scaling, feature selection, and tuning must respect the training-validation-test split.
- Gaussian squared losses are unbounded, so Lecture 2's bounded-loss test bound needs a different tail argument or a justified bounded loss.

A mathematically valid bound applies to a specified procedure and statistical model.

References for regularization

- **[RH]** Rigollet and Hütter, *High-Dimensional Statistics*, Sections 8–10 and 15–19.
  Author notes, arXiv:2310.19244
- **[RT]** Ryan Tibshirani, *Minimax theory*, *Ridge*, and *Lasso*, Berkeley Statistical Learning notes (2023).
  Course notes and sources
- **[RWY]** Raskutti, Wainwright, and Yu (2011), "Minimax rates of estimation for high-dimensional linear regression over ℓ_q-balls."
  Author manuscript, arXiv:0910.2042

The exact Gaussian risks and constants in the displayed lasso inequalities are derived using this lecture's own objective and RE normalizations.

Implementation references and reproducibility

Official estimator objectives

scikit-learn Ridge documentation

scikit-learn Lasso documentation

Supplied teaching package

Two student notebooks, a data dictionary, shared helpers, and fixed CSV splits. Separate instructor materials contain executable solutions, the generator, test labels, and the true support.

Reproducibility

The synthetic sample uses seed 20261015. No seed search or test-driven tuning was used. The package README records dependencies, penalty grids, and the fitting protocol.